

4.3 Chi-square tests

Further Probability and Statistics · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A chi-square goodness-of-fit statistic sums $(O-E)^2/E$ across categories. Expected counts must be large enough for approximation; pool categories where needed and adjust degrees of freedom for constraints/estimated parameters. Association tests use expected count row total $\times$ column total/grand total.

Worked example: follow the reasoning

For $O=12$ and $E=10$ in a category, contribution is $(12-10)^2/10=0.4$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Observed counts 30 and 20; expected 25 and 25. Calculate χ^2 and degrees of freedom for two categories with no fitted parameters. [3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

$$\chi^2 = (30-25)^2/25 + (20-25)^2/25 = 2.0 \text{ [2]}; \text{ df} = 2-1 = 1 \text{ [1]}.$$

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.

<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.